

$$y_{n+1} = \sqrt{2} y_n \quad \text{произв.}$$

$$g) \quad x_{n+1} = \frac{\cos y_n}{e^{x_n}} \quad \text{частн. произв.} < 1$$

$$y_{n+1} = \frac{2}{2}$$

Домашняя работа №2

$$E \approx \frac{h^2}{8} \max_{x \in [a, b]} |f''(x)| \Rightarrow h \leq \sqrt{\frac{8 \cdot 10^{-3}}{\max_{x \in [a, b]} |f''(x)|}}$$

$$a) \quad f''(x) = -\sin x \quad h \leq \sqrt{\frac{8 \cdot 10^{-3}}{1}} \approx 0,09$$

$$b) \quad f''(x) = 10 - \frac{1}{x^2} \quad h \leq \sqrt{\frac{8 \cdot 10^{-3}}{1}} \approx 0,09$$

$$b) \quad f''(x) = e^x \quad h \leq \sqrt{\frac{8 \cdot 10^{-3}}{2,72}} \approx 0,054$$

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \omega_{n+1}(x), \quad \omega_{n+1}(x) = (x-x_0)(x-x_1)\dots(x-x_n)$$

$$f^{(4)}(x) = \frac{24}{(A^2-x)^5}, \quad \text{возьмем max:} \quad \max_{[-4,7]} = \frac{24}{(A^2-1)^5}$$

$$\omega_4(x) = (x+4)(x+3)(x+2)(x+1)$$

$$\max_{[-4,7]} |\omega_4(x)| = |\omega_4(-2,5)| = 6,75$$

$$|R_3(x)| \leq \frac{\max_{[-4,7]} |f^{(4)}(x)|}{4!} \max_{[-4,7]} |\omega_4(x)| = \frac{6,75}{(A^2-1)^5} \leq 10^{-5}$$

$$A \geq 4,17$$

$$a) \quad x = \frac{x}{5} - \text{между } \frac{x}{6} \text{ и } \frac{x}{4}$$

$$x_0 = 0, \quad x_1 = \frac{x}{6}, \quad x_2 = \frac{x}{4}$$

$$R_2(x) = \frac{f'''(\xi)}{3!} (x-x_0)(x-x_1)(x-x_2), \quad f'''(x) = \sin x,$$

$$\max_{x \in [0, \frac{x}{4}]} |f'''(x)| = \frac{\sqrt{2}}{2} \left| \left(\frac{x}{5} - 0 \right) \left(\frac{x}{5} - \frac{x}{6} \right) \left(\frac{x}{5} - \frac{x}{4} \right) \right| \approx 0,0103$$

$$|R_2(x)| \leq \frac{1}{6} \cdot 0,0103 \cdot 0,4 = 1,2 \cdot 10^{-3}$$

$$\Delta = 1,2 \cdot 10^{-3} + 10^{-2} = 1,12 \cdot 10^{-2}$$

$$b) \quad x_0 = \frac{x}{6}, \quad x_1 = \frac{x}{4}, \quad x_2 = \frac{x}{3}$$

$$\max_{x \in [\frac{x}{6}, \frac{x}{3}]} |f'''(x)| = \frac{\sqrt{3}}{2} \approx 0,86$$

$$\left| \left(\frac{x}{24} - \frac{x}{6} \right) \left(\frac{x}{24} - \frac{x}{4} \right) \left(\frac{x}{24} - \frac{x}{3} \right) \right| \approx 0,0067$$

$$|R_2(x)| \leq \frac{1}{6} \cdot 0,0067 \cdot 0,86 \approx 9,6 \cdot 10^{-4}$$

$$\Delta = 10^{-4} \cdot 9,6 \cdot 10^{-4} = 9,6 \cdot 10^{-8}$$

N 6.4

$$I_{\text{sym}} = \frac{2^p I_{h/2} - I_h}{2^p - 1}$$

$$I = \int_a^b f(x) dx = I_T(h) = \frac{h}{2} \sum_{i=0}^n (f(x_i) + f(x_{i+1})),$$

$$I_T(h) = I + O(h^2) + O(h^4)$$

$$p=2 \quad I_{\text{sym}} = \frac{4(I + \frac{1}{4}h^2 + O(h^4)) - (I + Ch^2 + O(h^4))}{3} = \frac{3I + O(h^4)}{3} =$$

$$= I + O(h^4) \quad \text{ч.м.г.}$$

$$E = - \frac{(b-a)h^2}{12} f''(\xi), \quad b-a=1, \quad h = \frac{b-a}{n} = \frac{1}{n}$$

$$E \leq 10^{-4} \quad \frac{1}{12} \frac{1}{h^2} \max |f''(x)| \leq 10^{-4} \Rightarrow n \geq \frac{\max |f''(x)|}{12 \cdot 10^{-4}}$$

$$n \geq \frac{\sqrt{\max |f''(x)|}}{3,4 \cdot 10^{-2}}$$

$$a) \quad f''(x) = (4x^2 - 2)e^{-x^2}, \quad \max_b x \approx 0,62$$

$$\max |f''(x)| = 1,12 \quad n \geq \sqrt{933,33} \approx 30,57 \Rightarrow n \geq 31$$

$$b) \quad f''(x) = 2\cos(x^2) - 4x^2\sin(x^2), \quad \text{argmax} = 1$$

$$\max |f''(x)| = 2,5$$

$$n \geq \sqrt{2083,33} \approx 45,64 \Rightarrow n \geq 46$$

N 8.14(a)

$$\int_{-1}^1 f(x) dx \approx w_1 f(x_1) + w_2 f(x_2), \quad w_1 = w_2 = 1, \quad x_1 = -\frac{1}{\sqrt{3}}, \quad x_2 = \frac{1}{\sqrt{3}}$$

$$t = \frac{b-a}{2}x + \frac{a+b}{2} \Rightarrow \int_a^b f(x) dx \approx \frac{b-a}{2} \sum_{i=1}^2 w_i f(t_i)$$

$$x = \frac{1}{2}x + \frac{1}{2}, \quad dx = \frac{1}{2}dt$$

$$I = \frac{1}{2} \sum_{i=1}^2 \sin\left(\left(\frac{1+x_i}{2}\right)^2\right) \quad \frac{1 - \frac{1}{\sqrt{3}}}{2} = \frac{\sqrt{3}-1}{2\sqrt{3}} \quad \frac{1 + \frac{1}{\sqrt{3}}}{2} = \frac{\sqrt{3}+1}{2\sqrt{3}}$$

$$I = \frac{1}{2} \sin\left(\frac{(\sqrt{3}-1)^2}{2\sqrt{3}}\right) + \frac{1}{2} \sin\left(\frac{(\sqrt{3}+1)^2}{2\sqrt{3}}\right) = \frac{1}{2} \cdot 0,8 \cdot 10^{-2} + \frac{1}{2} \cdot 0,01 =$$

$$= 5,4 \cdot 10^{-3}$$

$$E = \frac{(b-a)^{2n+2}}{(2n+2)!} \max |f^{(2n+2)}(\xi)| \quad E \leq \frac{1}{120 \cdot 5} \cdot \frac{40}{1} = \frac{1}{600} \cdot 40 = \frac{1}{15}$$

$I(f) = \int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx$ №8.12 (б) — чебышев. многочлен 1-го рода

$$g_k = \cos\left(\frac{(2k-1)\pi}{2n}\right), \quad k=1, 2$$

$$x_1 = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}, \quad x_2 = \cos \left(\frac{3\pi}{4} \right) = -\frac{\sqrt{2}}{2}$$

$$W_k = \frac{P}{n} = \frac{80}{2}$$

$$I(f) = \sum_{k=1}^2 w_k f(x_k) = \frac{\sqrt{2}}{2} \left[f\left(\frac{\sqrt{2}}{2}\right) + f\left(-\frac{\sqrt{2}}{2}\right) \right]$$

$$\int_a^\infty f(x) dx = \int_a^A f(x) dx + \int_A^\infty f(x) dx, \quad \int_A^\infty f(x) dx < \varepsilon$$

$$\text{bvc)} \int_1^{\infty} \frac{1}{x^2} \sin \frac{1}{x^2} dx, \quad \int_A^{\infty} \frac{1}{x^2} \sin \frac{1}{x^2} dx \sim \int_A^{\infty} \frac{1}{x^4} dx = -\frac{1}{3x^3} \Big|_A^{\infty} =$$

$$= \frac{1}{3A^3} < 10^{-4} \Rightarrow A \geq 14,7 \quad A = 15 \text{ wählen}$$

$$\textcircled{d} \quad \epsilon = \frac{(b-a)h^2}{12} \quad \max |f'(x)| = \frac{14}{12} h^2 \cdot 0,02 \leq 10^{-4} \Rightarrow h \leq 0,021$$

8c) $\int_0^1 \frac{dx}{\sqrt{2(2x-1)^2}}$, $x = \cosh^2 t$, $dx = 2 \sinh t \cosh t$

$$\sqrt{a(\cosh)} = \sqrt{\sinh^2 t (\sinh^2 t + 1)} = \sinh t \cosh t$$

$$I = \int_0^{\sinh^{-1}(1)} \frac{2 \sinh t \cosh t}{\sinh t \cosh t} dt = 2 \int_0^{\sinh^{-1}(1)} dt = 2 \sinh^{-1}(1) \approx 0,99$$

$$f''(x) = 2, \quad \frac{h^2}{12} \times 2 \leq 10^{-3} \quad h \leq 0,077$$

$$y' = ay, \quad y(0) = 1 \quad y(t) = e^{at}$$

Имя ^{на} ~~на~~ савва Фидера!

$$y_{n+1} = y_n + h a y_n, \quad y_{n+1} = y_n (1 + h a), \quad y_n = (1 + h a)^n y_0,$$

$$y_0 = y(0) = 1 \Rightarrow y_n = (1 + ha)^n$$

Следовательно: $\lim_{h \rightarrow 0} \max_n |y_n - y(t_n)| = 0$

$$y(t_n) = e^{at_n} = \sum_{k=0}^{\infty} \frac{(at_n)^k}{k!} = \sum_{k=0}^{\infty} \frac{(a n h)^k}{k!}$$

$h \rightarrow 0 \Rightarrow y_n \rightarrow 0, e^{a b n} \rightarrow 0 \Rightarrow \text{сходим. с.м.с. } O(h)$

Решая: $y_{n+1} = y_n + ha y_{n+1}$, $y_{n+1} = \frac{y_n}{1-ha}$, $y_n = \frac{y_0}{(1-ha)^n}$
 $y_n = (1-ha)^{-n}$

$$(1-ha)^{-n} = \sum_{k=0}^{\infty} C_n^k (ha)^k \Rightarrow O(h) \text{ с точностью.}$$

Метод Рунге-Кутты: $y_{n+1} = y_n + \frac{h}{2} a(y_n + y_{n+1})$

$$y_{n+1} = \frac{1 + \frac{ha}{2}}{1 - \frac{ha}{2}} y_n, \quad y_n = \left(\frac{1 + \frac{ha}{2}}{1 - \frac{ha}{2}} \right)^n = (1 + ha + O(h^2))^n$$

$h \rightarrow 0$ с точностью $O(h)$

№ 7.5

$$p=1 \quad \sum b_i = 1$$

$$p=2 \quad \sum b_i c_i = \frac{1}{2}$$

$$p=3 \quad \sum b_i c_i^2 = \frac{1}{3}, \quad \sum b_i \sum a_{ij} c_j = \frac{1}{6}$$

№ 7.10

$$y_{n+1} = y_n + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = f(t_n, y_n)$$

$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2} k_1\right) \quad k_4 = f(t_n + h, y_n + k_3 h)$$

$$k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2} k_2\right)$$

Самуэлсон: $\int_a^b f(x) dx = \frac{h}{6} [f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)]$, $h = \frac{b-a}{2}$

$$\frac{dy}{dt} = f(t, y) \Rightarrow y_{n+1} = y_n + \int_{t_n}^{t_{n+1}} f(t, y) dt = y_n + \frac{h}{6} [f(t, y) + 4f\left(t_n + \frac{h}{2}, y_n\right) + f(t_n + h, y)] \sim \text{Самуэлсон}$$

№ 9.8

$$y_{l+1} = y(x_l + h) = y(x_l) + hy'(x_l) + \frac{h^2}{2} y''(x_l) + \frac{h^3}{6} y'''(x_l) + O(h^4)$$

$$y_{l-1} = y(x_l - h) = y(x_l) - hy'(x_l) + \frac{h^2}{2} y''(x_l) - \frac{h^3}{6} y'''(x_l) + O(h^4)$$

$$y' \approx \frac{y_{l+1} - y_{l-1}}{2h} = y' + \frac{h^2}{6} y''' + O(h^4) \Rightarrow \text{точность } O(h^2)$$

$$y'' \approx \frac{y_{l+1} - 2y_l + y_{l-1}}{h^2} = y'' + O(h^2) = y'' + O(h^2) \Rightarrow O(h^2)$$

$\Rightarrow$ а разность между y'' и y'' - $O(h^2)$

№ 9.13

$$\sum b_i = 1, \quad \sum b_i c_i = \frac{1}{2}, \quad \sum b_i c_i^2 = \frac{1}{3}, \quad \sum b_i \sum a_{ij} c_j = \frac{1}{6}$$

$$\sum b_i = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1 \checkmark$$

$$\sum b_i c_i = \frac{1}{6} \cdot 0 + 1 \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{2} + \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{2} \checkmark$$

$$\sum b_i c_i^2 = \frac{1}{6} \cdot 0 + \frac{1}{6} \cdot 1 + \frac{4}{6} \cdot \frac{1}{4} = \frac{1}{3} \checkmark$$

$$\sum b_i \sum a_{ij} c_j = \frac{1}{6} \cdot 0 + \frac{1}{6} \cdot (1+0) + \frac{4}{6} \cdot \frac{1}{2} = \frac{1}{3} \neq \frac{1}{6}$$

гробавиш $a_{33} = -\frac{1}{3}$, мого $\sum c_i = \frac{1}{3}$

№9.16

$$\frac{dy}{dx} + y = x^2 + 2x$$

$$y_{m-1} = y_m - h y_m + \frac{h^2}{2} y_m - \frac{h^3}{6} y_m + O(h^4)$$

$$y_{m-2} = y_m - 2h y_m + 2h^2 y_m - \frac{4}{3} h^3 y_m + O(h^4)$$

Тече упрошениа:

$$\frac{2h y_m - \frac{2}{3} h^3 y_m}{2h} + y_m = y_m - \frac{h^2}{3} y_m + y_m = 2y_m - \frac{h^2}{3} y_m = x_m^2 + 2x_m$$

Ошибка апроксим. $-\frac{h^2}{3} y_m = O(h^2)$ га, апроксим.

№10.3

$$a) \frac{d^2 y}{dt^2} = y \sin t$$

$$y(0) = 0, y'(0) = 1, t \in [0, \pi]$$

$$a) u(t) = y(t), v(t) = y(t) \quad \begin{cases} \dot{y} = x - \text{замена} \\ \dot{x} = y \sin t \end{cases}$$

$$b) y''(t_n) = \frac{y_{n+1} - 2y_n + y_{n-1}}{h^2} = y_n \sin t_n$$

$$y_{n+1} = h^2 y_n \sin t_n + 2y_n - y_{n-1}, y_0 = 0, y_1 = h - \text{беруа}$$